PROJECT PROPOSAL

REARGUARD AI

Intelligent Rear Collision Prediction & Warning System

> Detect Early. Warn Smart. Ride Safe.

PROBLEM & SOLUTION

The Road Safety Challenge

The Hazard

Riders face extreme risk from high-speed rear-approaching vehicles due to limited rear awareness and delayed human reaction times.

The Proposed Solution

A smart two-wheeler system that actively monitors rear traffic and issues early collision warnings before impact.

ARCHITECTURE

Core Functions

1. Detect

Rear-facing sensor detects vehicles behind.

2. Calculate

Estimates distance and relative closing speed.

3. Predict

Intelligent algorithm evaluates collision likelihood.

4. Alert

Triggers audio and visual signals to rider.

DETECTION LOGIC

Time to Collision (TTC)

CORE EQUATION

TTC = Distance / Closing Speed

Alarms trigger on risk dynamics rather than fixed distance thresholds alone.

Example Scenario

Distance: 60 metres

Closing Speed: 20 m/s

TTC: 3.0 Seconds (Dangerous)

ALERT MECHANICS

Smart Warning Logic

Risk Level	System Response
Low	Monitor silently
Medium	Yellow visual warning
High	Red warning + audible alarm

Hysteresis Integration

Prevents annoying rapid toggling. Warning clears only when risk drops below a secondary lower threshold for a set duration.

ROAD DYNAMICS

Context-Aware Guidance

Narrow Roads

Warns rider to maintain a steady path without sudden moves.

Single-Lane Roads

Gives early warnings without suggesting unsafe escapes.

Multi-Lane Highways

Recommends escape routes only when clear space is confirmed.

Core Principle: If no safe path is verified, no direction is recommended.

TECHNOLOGY STACK

Enabling Technical Components

Camera / Sensor

Rear-mounted optic capturing real-time traffic visuals behind the vehicle.

Radar / LiDAR

Measures exact distance and relative approach speed with minimal delay.

AI Processing Unit

Analyzes sensor streams instantly to calculate collision threats and trajectories.

Display + Alarm

Compact visual display paired with audio alerts for direct rider feedback.

VALUE PROPOSITION

Key Innovation Value

Proactive Security

Proactive warning system rather than passive impact recording.

Smart Filter Logic

Filters out irrelevant traffic to significantly reduce false alarms.

Road-Adaptive Decisions

Adapts alerting logic specifically for narrow roads and multi-lane highways.

CORE INNOVATION

Intelligent Context-Aware Protection

Reduced False Alarms

Dynamic risk filtering based on closing speed to prevent alert fatigue.

Adaptive Road Geometry

Contextual awareness tailored for both narrow city streets and multi-lane highways.

Safe Route Verification

Evaluates available clearance space before suggesting rider evasive action.

TECHNICAL ARCHITECTURE

Engineered for Technical Credibility

01. DYNAMIC DISTANCE

Adaptive Warning Zone

Warning boundaries calculate dynamically using relative speed and reaction window rather than fixed 80–100m thresholds.

02. ESCAPE LOGIC

Fail-Safe Warning Fallback

Issues general alerts when side-view clearance is unverified, avoiding false direction guidance in narrow terrain.

03. SENSOR INTEGRATION

Multi-Modal Radar + Vision

Combines radar distance metrics with camera imagery to mitigate environmental interference like rain or sharp bends.

TECHNICAL DEFENSE

Judge Q&A: System Fundamentals

Q1

Why not simply rely on rear-view mirrors?

Mirrors require active rider focus and manual calculation. REARGUARD provides continuous automated trajectory monitoring and proactive hazard alerts.

Q2

How is approaching speed accurately detected?

Rear millimeter-wave radar continuously tracks distance and Doppler frequency shifts to compute real-time closing speed and collision window.

TECHNICAL DEFENSE

Judge Q&A: Edge Cases & Scope

Q3 **What happens on narrow roads?**

When lateral escape paths are constrained, directional prompts disengage and the system emits a primary visual/audible hazard warning.

Q4 **How are nuisance alarms avoided?**

Alerts trigger strictly when calculated Time-to-Collision (TTC) crosses safe safety thresholds, ignoring stationary or parallel traffic.

Q5 **Does this guarantee complete accident prevention?**

No. It functions strictly as an Advanced Rider Assistance System (ARAS) designed to augment rider reaction time and situation awareness.



"Most safety systems warn us about what is happening around us. REARGUARD AI aims to warn us about what could happen next."